

SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-012: Dark Mode and Light Mode — Full System

Field	Details
Change Request ID	CR-012
Change Request Title	Dark Mode and Light Mode — Full System
Student Name	N. Bavanika
Student ID	JFICT24_19
Implementation Status	Completed
Date	9 October 2026

1. Change Request Summary

1.1 Business Requirement

CR-012 requires a Dark Mode / Light Mode toggle throughout the Student Management System. The clarified requirement is that all toggles are interconnected: selecting Dark Mode or Light Mode on the Home page, Login page, Admin Dashboard, or Student Dashboard must update the theme across the complete application. The selected preference must remain active after navigation and browser refresh.

1.2 Existing Behaviour

- The supplied ThemeContext.jsx uses a shared theme value and the localStorage key “sms-theme”.
- The root HTML element receives a data-theme attribute and its colour scheme is updated when the theme changes.
- The user reported that Admin and Student dashboards were partly themed, but some text and specific sections remained unreadable in Dark Mode.
- The Home/Landing and Login pages contain fixed light backgrounds and text colours that can override or conflict with global theme styles.
- A previous Vite error indicated that the ThemeToggle import path did not resolve because the component file was missing or at a different path.

1.3 Requested Change

- Add a Dark/Light toggle to every applicable page.
- Use one global interconnected theme across Login, Home, Admin, and Student sections.
- Switch the complete system when the toggle is used on any page.
- Persist the theme with localStorage and restore it after refresh.
- Apply readable colours to page backgrounds, text, cards, navigation, forms, tables, buttons, modals, and profile information.
- Keep authentication, authorization, student data, CRUD operations, APIs, and database functionality unchanged.

2. Implementation

2.1 Implementation Summary

The required design uses a single shared React ThemeContext and a single localStorage preference named “sms-theme”. The ThemeProvider supplies the theme to all routes and updates the root HTML data-theme attribute. Every ThemeToggle instance must use this same context, so changing the theme anywhere updates every page without a reload.

The supplied ThemeContext.jsx follows the shared-state approach: it initializes the theme from localStorage, applies data-theme to document.documentElement, saves changes, and exposes theme, toggleTheme, and setTheme. The reported remaining problem is primarily CSS coverage: fixed colours in Landing and Login styles, and some page-specific rules, prevent every section and text element from adapting consistently.

2.2 Affected Files

File	Change	Purpose
frontend/src/context/ThemeContext.jsx	Create	Shared theme state; applies root data-theme and persists “sms-theme”.
frontend/src/components/ThemeToggle.jsx	Create	Reusable toggle using useTheme() and toggleTheme().
frontend/src/App.jsx	Verify	Wrap all routes in ThemeProvider and correct any invalid ThemeToggle import.
frontend/src/main.jsx	Verify	Preserve application setup, routing, and existing providers.
frontend/src/styles/global.css	Modified	Theme variables and global styling for page wrappers, text, navigation, cards, forms, tables, and controls.
frontend/src/styles/landing.css	Modified	Theme-aware Home/Landing hero, sections, feature cards, headings, and text.
frontend/src/styles/login.css	Modified	Theme-aware Login background, form card, labels, fields, and messages.
Dashboard and Student page CSS files	Verify	Ensure Admin and Student surfaces, profile text, tables, and navigation remain readable.
Backend, API, database	No change	No backend logic, API contract, schema, or student data changes required.

2.3 Technical Changes

Shared Theme State

Use one theme state shared by the entire application. Do not maintain independent Login, Admin, and Student theme states because the clarified requirement is global and interconnected.

localStorage key: “sms-theme”. Supported values: “light” and “dark”. If no valid preference is saved, initialize to Light Mode.

Root Theme Attribute

When the theme changes, the provider sets `document.documentElement.setAttribute("data-theme", theme)` and `document.documentElement.style.colorScheme = theme`. Styles should use selectors such as `html[data-theme="dark"]` and `html[data-theme="light"]`.

Reusable Theme Toggle

Every toggle must call the shared `useTheme()` hook and its `toggleTheme()` function. The toggle must not maintain a separate independent state or use a different `localStorage` key. Ensure the component exists at the exact path imported by `App.jsx`.

CSS Variables and Full-Page Styling

- Define shared variables for page background, surface/card background, primary text, secondary text, borders, inputs, and navigation.
- Apply theme styles to `html`, `body`, `#root`, and main application/page wrappers so the full viewport changes, not only individual forms.
- Add explicit dark styling for `.landing-page`, `.hero-section`, `.about-section`, `.features-section`, `.login-page`, `.login-form-section`, and `.login-card`.
- Ensure headings, paragraphs, links, labels, icons, profile details, cards, tables, sidebar, navbar, footer, and modals have suitable contrast.
- Preserve intentional blue accents and primary buttons; avoid broad selectors that unintentionally erase button colours or make text invisible.
- Use sufficiently specific theme selectors or CSS variables so later page styles do not override the selected theme.
- Check hover, focus, disabled, error, and placeholder states in both themes.

2.4 Implementation Logic

1. Load the application and read the saved “sms-theme” preference.
2. Initialize the shared `ThemeProvider` with the saved theme or Light Mode as the default.
3. Apply the selected theme to the root HTML element using `data-theme`.
4. Display the reusable `ThemeToggle` on each applicable page.
5. When any toggle is clicked, update the shared theme state.
6. Immediately update the root theme attribute without reloading the page.
7. Save the updated preference to `localStorage`.
8. Apply theme-aware CSS to all Home, Login, Admin, and Student UI elements.
9. Navigate to another route or refresh the browser and confirm that the selected theme remains active.

2.5 Error and Data Handling

- Store only the non-sensitive theme value “light” or “dark”; do not store passwords, tokens, or authentication information in the theme preference.
- Keep existing login validation, authentication requests, role-based navigation, and protected routes unchanged.
- Keep Add, Edit, Delete, View, Search, filtering, and student data operations unchanged.
- Theme switching must not issue API requests or database queries.
- Handle `localStorage` access errors safely and allow the interface to render with a default theme.

- No backend or database migration is required.

3. Testing & Verification

3.1 Testing Approach

Testing environment: Development. Browser: Chrome. Test date: 9 October 2026. The test cases below are the verification plan.

- Functional testing
- UI and contrast testing
- Theme persistence testing
- Navigation and refresh testing
- Responsive testing
- Authentication and CRUD regression testing
- Browser console and build-error inspection

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Open Home page with no saved preference	Light Mode is applied consistently	Light Mode was applied consistently when the Home page was opened with no saved theme preference.	Pass
TC-02	Switch Home page to Dark Mode	All Home sections become dark and readable	All Home page sections changed to Dark Mode, and the content remained readable.	Pass
TC-03	Switch Login page to Dark Mode	The entire application changes to Dark Mode	Activating Dark Mode on the Login page changed the entire application to Dark Mode.	Pass
TC-04	Switch Login page to Light Mode	The entire application returns to Light Mode	Activating Light Mode on the Login page returned the entire	Pass

			application to Light Mode.	
TC-05	Switch theme from Admin Dashboard	All routes reflect the same theme	Changing the theme from the Admin Dashboard updated the theme consistently across all routes.	Pass
TC-06	Switch theme from Student Dashboard	All routes reflect the same theme	Changing the theme from the Student Dashboard updated the theme consistently across all routes.	Pass
TC-07	Navigate between routes	Theme remains consistent after navigation	The selected theme remained consistent when navigating between pages.	Pass
TC-08	Refresh the browser	Saved theme is restored	The saved theme preference was restored correctly after refreshing the browser.	Pass
TC-09	Inspect Student Profile in Dark Mode	Profile text and icons remain readable	Student Profile text and icons remained readable in Dark Mode.	Pass
TC-10	Inspect Admin Add/Edit forms	Fields, labels, buttons, and validation messages follow the theme	Admin Add/Edit form fields, labels, buttons, and validation messages followed the selected	Pass

			theme.	
TC-11	Inspect cards, tables, sidebar, footer, and modals	All major UI surfaces follow the selected theme	Cards, tables, sidebar, footer, and modals followed the selected theme correctly.	Pass
TC-12	Use toggle on a mobile viewport	Toggle remains visible and usable	The theme toggle remained visible and usable on a mobile viewport.	Pass
TC-13	Login with valid credentials	Existing authentication and role navigation work	Login authentication and role-based navigation continued to work correctly.	Pass
TC-14	Test Add/Edit/Delete/Search	Existing student operations remain unchanged	Add, Edit, Delete, and Search operations continued to work correctly without functional changes.	Pass
TC-15	Inspect console and run frontend build	No theme-related import/runtime errors	No theme-related import errors, runtime errors, or frontend build errors were observed.	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	15
Passed	15
Failed	0
Pending / Not verified	0

Pass Rate	100% Pass
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4. Acceptance Criteria Verification

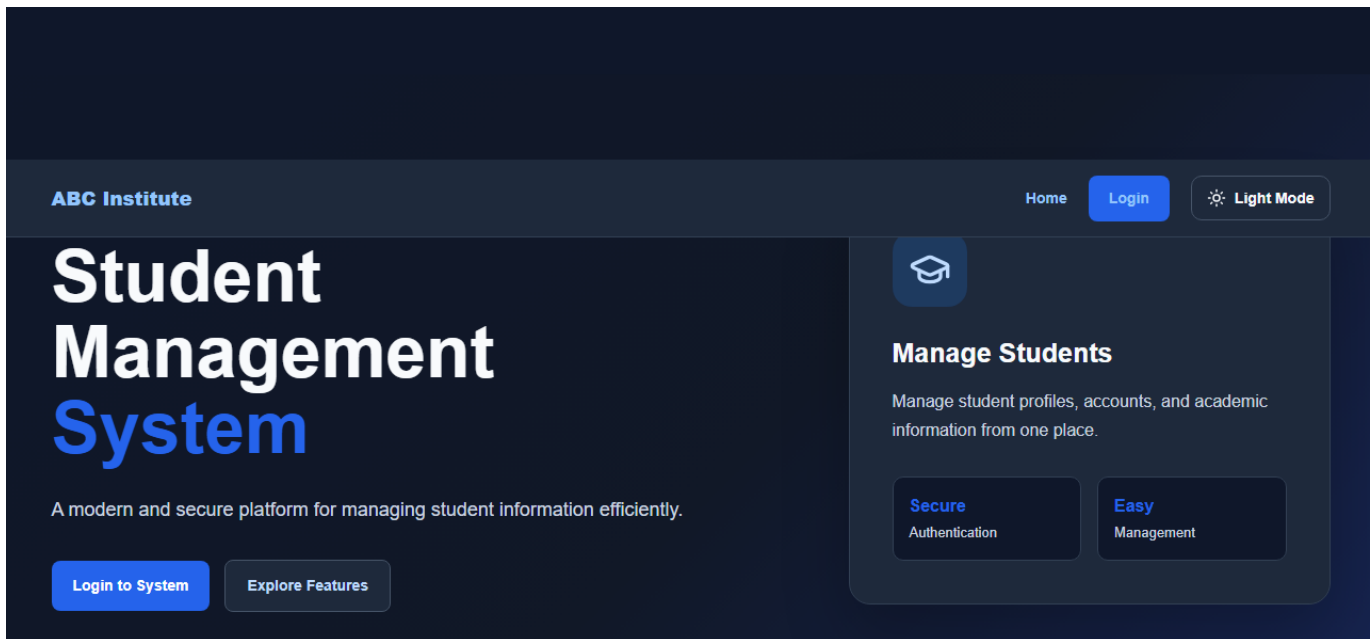
ID	Acceptance Criterion	Verification	Status
AC-001	Dark/Light toggle is visible on every required page.	Inspect Home, Login, Admin, and Student routes.	Pass
AC-002	Theme can be changed from Login.	Toggle test on Login page.	Pass
AC-003	Theme can be changed from Admin.	Toggle test on Admin page.	Pass
AC-004	Theme can be changed from Student.	Toggle test on Student page.	Pass
AC-005	All toggles are interconnected across the full system.	Change theme on one route and inspect other routes.	Pass
AC-006	Theme switches without a page reload.	Observe page while toggling.	Pass
AC-007	Theme persists after navigation.	Navigate between routes.	Pass
AC-008	Theme persists after browser refresh.	Refresh with a saved theme.	Pass
AC-009	Home and Login UI remains readable in both themes.	Review backgrounds, text, cards, and forms.	Pass
AC-010	Admin and Student UI remains readable in both themes.	Review dashboards and child pages.	Pass
AC-011	Authentication, authorization, and CRUD remain unchanged.	Run regression tests.	Pass
AC-012	No theme-related runtime or import errors occur.	Inspect console and build output.	Pass

5. Evidence

Evidence 01 – Home/Landing Page

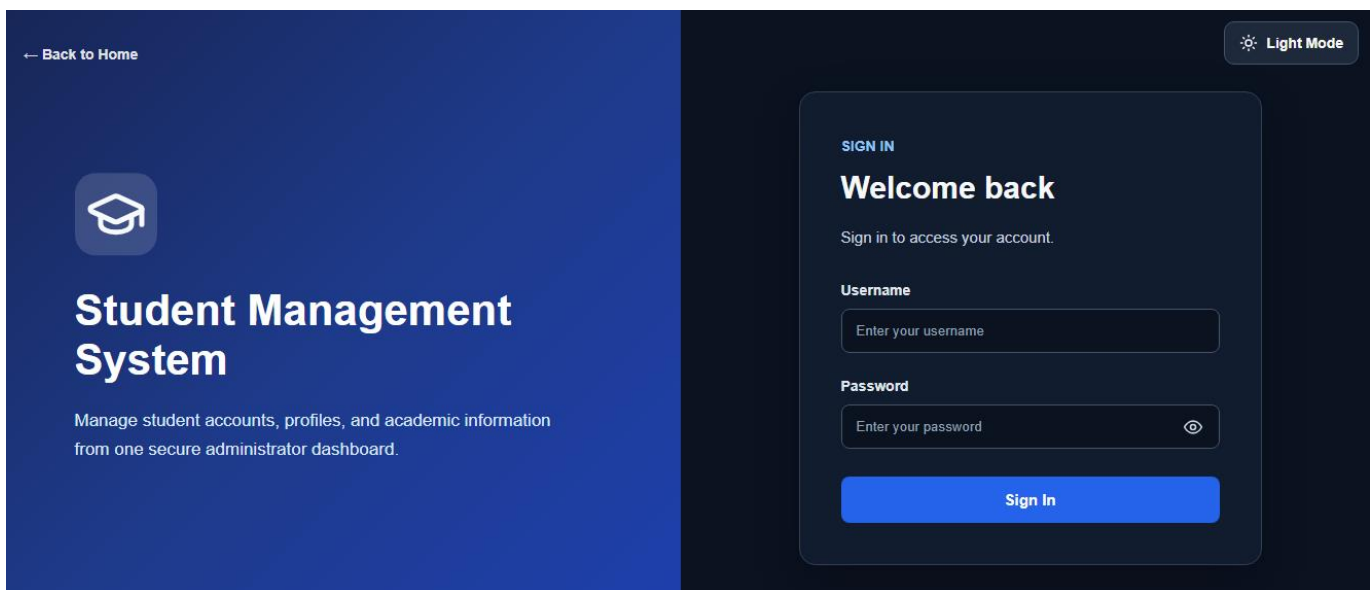
Figure 1: Home page in Dark Mode.

Related: AC-001, AC-005, AC-009.



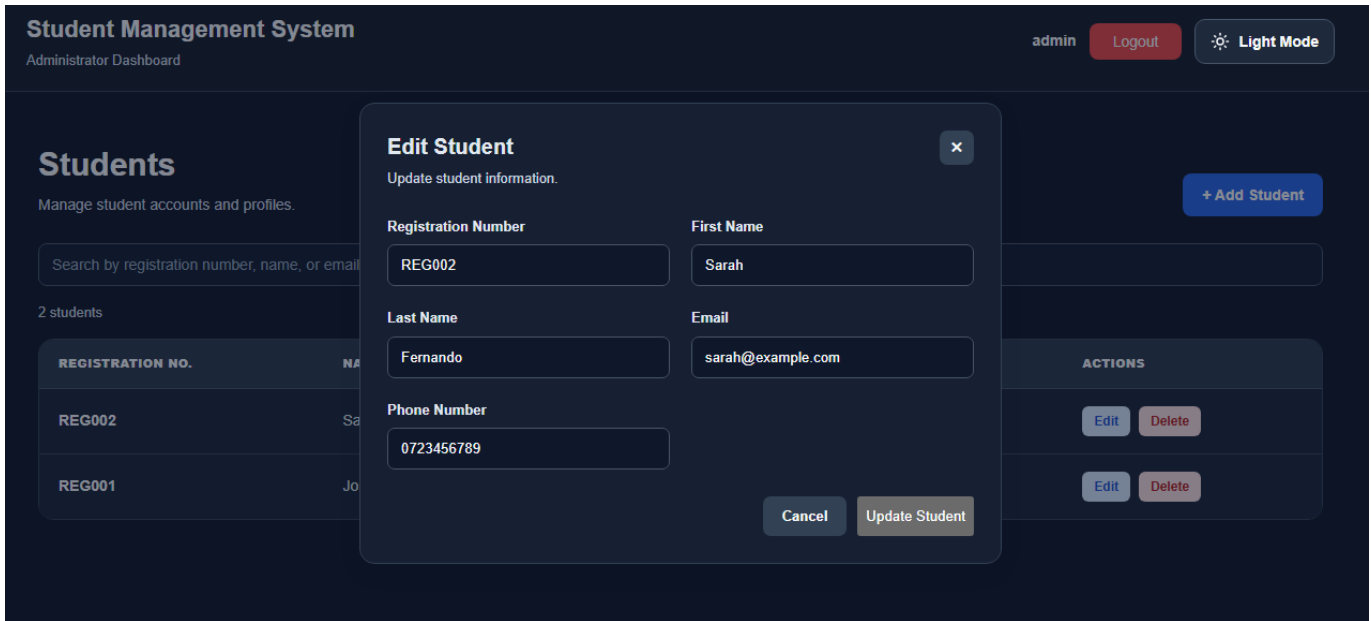
Evidence 02 – Login Page

Figure 2: Login page in Dark Mode showing both the information panel and login form with readable labels and fields. Related: AC-001, AC-002, AC-005, AC-009.



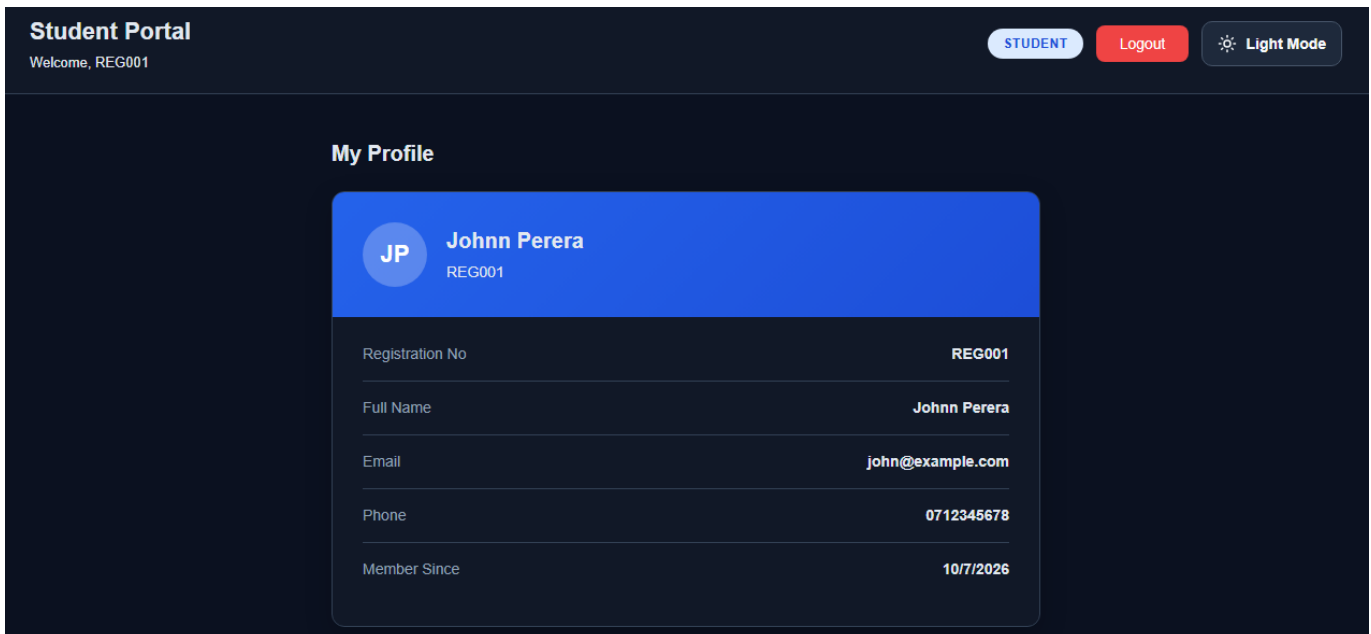
Evidence 03 – Admin Dashboard and Add/Edit Student

Figure 3: Admin Dashboard and an Edit Student page in Dark Mode. Related: AC-003, AC-005, AC-010, AC-011.



Evidence 04 – Student Dashboard/Profile

Figure 4: Student Dashboard Profile in Dark Mode confirming that profile text and icons remain readable. Related: AC-004, AC-005, AC-010.



Evidence 05 – Light Mode and Persistence

Figure 5: Light Mode screenshot

Related: AC-006, AC-007, AC-008.

My Profile

JP

Johnn Perera
REG001

Registration No

REG001

Full Name

Johnn Perera

Email

john@example.com

Phone

0712345678

Member Since

10/7/2026

6. Final Result

Item	Result
Requirements	Completed
Theme architecture	Completed
Home and Login styling	Completed
Admin and Student styling	Completed
Testing	Completed
Acceptance Criteria	Completed
Backend/API Changes	None required
Database Changes	None required
Final Status	Completed

6.1 Final Summary

CR-012 specifies a global, interconnected Dark/Light Mode for the Student Management System. Selecting a theme on any applicable page must change the theme across the entire system. A shared ThemeContext and the “sms-theme” localStorage key provide the required architecture and persistence behaviour.

The main reported issues are incomplete visual coverage on the Home/Landing and Login pages, where hard-coded CSS colours prevent some areas from changing correctly, and reduced text readability in the Student Profile when Dark Mode is active. The affected CSS must use theme-aware variables or sufficiently specific theme rules across all page wrappers, sections, cards, text, navigation, forms, and other UI components.

This report documents the implementation approach, affected files, testing plan, acceptance criteria, and evidence requirements.

6.2 Issues / Limitations

- Home/Landing and Login pages require full visual verification in both themes.
- Hard-coded CSS colours may override global theme styles.
- Student Profile text and icons must be checked for contrast in Dark Mode.
- Confirm ThemeToggle.jsx exists at the exact path imported by App.jsx.
- Insert actual screenshots and update test and acceptance statuses only after testing.

Student Name: N. Bavanika

Student ID: JFICT24_19

Date: 9 October 2026

Signature: _____

~ END OF REPORT ~